

2. Distinct values of $\left\lfloor \frac{n}{i} \right\rfloor$

Now, let's get to even more interesting facts:

Fact 2

Over all positive integers i , there are $O(\sqrt{n})$ distinct values of $\left\lfloor \frac{n}{i} \right\rfloor$.

We will discuss why this is true and how to retrieve the exact values over the exact ranges in $O(\sqrt{n})$.

Proof of Fact 2

We note that for $i > \sqrt{n}$, we have that $\left\lfloor \frac{n}{i} \right\rfloor \leq \frac{n}{i} < \frac{n}{\sqrt{n}} = \sqrt{n}$ and there are obviously at most $\sqrt{n}$ values for $i \leq \sqrt{n}$, this makes a total of at most $2\sqrt{n} \in O(\sqrt{n})$.

Now, note that these values are non-increasing as i increases. For example, for $n = 10$, we get the values (starting from $i = 1$) 10, 5, 3, 2, 2, 1, 1, 1, 1, 0, 0, ... So, to find these values, we will prove the following fact:

Fact 3

Let $S = \left\{ \left\lfloor \frac{n}{i} \right\rfloor : i \in \mathbb{Z}^+ \right\}$ — in other words, the set of all distinct values of $\left\lfloor \frac{n}{i} \right\rfloor$. Let $q \in S$ where $q > 0$ and let l be the minimum positive integer such that $\left\lfloor \frac{n}{l} \right\rfloor = q$ and let r be the maximum positive integer such that $\left\lfloor \frac{n}{r} \right\rfloor = q$, then

$$r = \left\lfloor \frac{n}{\left\lfloor \frac{n}{l} \right\rfloor} \right\rfloor$$